

LANGUAGES AND TOOLS

Every website basically has two parts – the front end (all the parts seen in the browser that users interact with) and the back end (the part that powers the front end, which deals with storing data), resulting in three main categorisations of web development jobs: front end development, backend development, and full-stack development.

Front-end development uses HTML (building the structural blocks), CSS (adding styling and visual effects) and JavaScript (adding functionality and allowing interaction (behaviour) to the page). HTML is a markup language, CSS is a styling language, and JavaScript is a programming language.

Frameworks for Front-End Development

They help front-end development handle repetitive tasks. Popular ones include React (technically a library), Angular and Vue. They help build applications faster.

Version Control

This help keeps track of project history and collaborate with other programmers. Git is very popular for this.

HTML – HYPERTEXT MARKUP LANGUAGE

It is a markup language that defines the structure of a webpage. Your web browser interprets it to display content on your screen.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Hello!</title>
  </head>
  <body>
    Hello, world!
  </body>
</html>
```

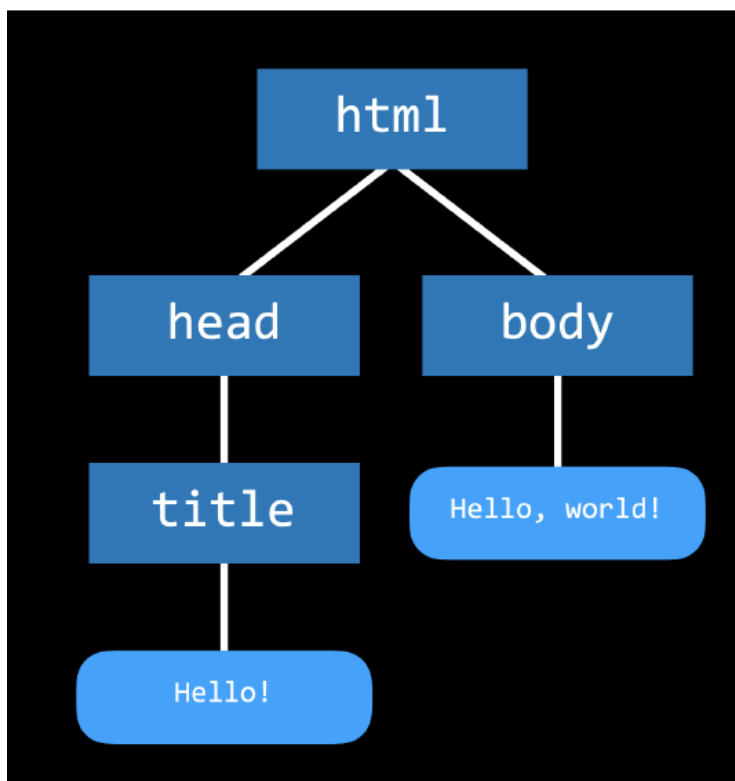
This code can be explained as follows:

- In the first line, **<!DOCTYPE html>** we are declaring (to the web browser) that we are writing the document in HTML
- After that, the page consists of nested **HTML elements** (such as html and body), each with an **opening and closing tag** marked with either **<element>** for an opening and **</element>** for a closing. The **<html>** element is the root element of the HTML page.
- Notice how each inner element is indented further than the last. While the browser does not necessarily require this, keeping this up in your code will be very helpful.

- HTML elements can include **attributes**, which give the browser extra information about the element. For example, when we include lang="en" in our initial tag, we tell the browser we are using English as our primary language.
- Inside the HTML element, we typically want to include a head and a body tag. The head element <head> will include information about your page that is not necessarily displayed, (meta information), and the body element <body> will contain what is visible to users who visit the site – text, headings, images, tables, hyperlinks, lists, etc.
- Within the head, we have included a title for our webpage, which you'll notice is displayed in the tab at the top of our web browser.
- Finally, we've included the text "Hello, world!" in the body, which is the visible part of our page.

DOCUMENT OBJECT MODEL (DOM)

The DOM is a convenient way of visualising how HTML elements relate to each other in a tree-like structure. Above is an example of the DOM layout for the page we just wrote.



MORE HTML ELEMENTS

In many cases, an element consists of a tag, some content, and a closing tag. Not all tags have both an opening and a closing tag, and these are called empty tags. We will see more examples shortly. You may want to use many HTML elements to customise your page, including headings, lists, and bolded sections. In the following example, we'll see a few of these in action. One more thing to note: <!-- --> gives us a comment in HTML, so we'll use that below to explain some of the elements.

HTML HEADINGS

HTML headings are defined with the <h1> to <h6> tags. <h1> defines the most important (or largest) heading while <h6> defines the least important (or smallest) heading.

Example:

```
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
<h4>This is heading 4</h4>
<h5>This is heading 5</h5>
<h6>This is heading 6</h6>
```

PARAGRAPHS

A paragraph is defined with the <p> tag:

```
<body>
  <p>This is my first paragraph</p>
  <p>And this is my second paragraph</p>
</body>
```

LINKS

Links (hyperlinks) are defined using the <a> tag. The link's destination (URL) is specified in the **href** attribute. The item that will be displayed for the user to click is specified after the opening tag and before the closing tag. Attributes provide additional information about HTML elements.

```
<body>
  <a href="https://www.knust.edu.gh">KNUST</a>
</body>
```

!!! Students to read:

- **How to open a link in a new browser window or tab.**
- **Linking to an email address**
- **Absolute URLs and Relative URLs**
- **Link titles**

IMAGES

Images are defined with the **** tag. The source file (**src**), alternative text (**alt**), **width**, and **height** are provided as attributes. The width and height attributes (specified in pixels) help size the image being added to the page and are, therefore, optional but recommended. The alternative will be displayed in cases where the object cannot be loaded for a user. **Note:** The **** tag is an empty tag – it does not have an opening and a closing tag.

```
<body>
  
</body>
```

!!! Students to read:

- Using an image as a link
- Floating images
- Image maps
- Using background images

OTHER TAGS

**** - Bolden text

<i> - To italicise text

**
** - Introduces a line break or a new line in the script

<hr> - Displays a thematic change in the content

<blockquote> - Displays text as a block 40 pixels from the left border

<pre> - This defines a preformatted text.

```
<body>
  <b>This text is bold</b>
  <i>This is italicised</i>
  <br>
  <hr>
  <blockquote>Mary had a little lamb
    it's fleece was white as snow
    and everywhere that Mary went
    the lamb was sure to go
  </blockquote>
  <hr>
```

```
<pre>
  Mary had a little lamb
  it's fleece was white as snow
  and everywhere that Mary went
  the lamb was sure to go
</pre>
</body>
```

MORE ELEMENTS (DISCUSS)

- **** - Important text
- **** - Emphasized text
- **<mark>** - Marked text
- **<small>** - Smaller text
- **** - Deleted text
- **<ins>** - Inserted text
- **<sub>** - Subscript text
- **<sup>** - Superscript text
- **<q>**
- **<abbr>**
- **<address>**
- **<cite>**
- **<bdo>**

STYLES

The **style** attribute adds styles, such as colour, font size, and more, to elements. The syntax is
`<tag style="property:value;">`

Some common styles include background-color, text color, fonts, text size and text alignment.

```
<body>
  <h1 style="background-color: green;">My heading with a green background</h1>
  <p style="color:red;">This is a paragraph with red text</p>
  <p style="font-family: verdana;">This is in Verdana font style</p>
  <p style="font-size: 400%;">This is a styled paragraph</p>
  <h3 style="text-align: center;">This heading is centered</h3>
</body>
```

COLOURS

In HTML, colours are specified with predefined names, or with RGB, HEX, HSL, RGBA or HSLA values.

Check out: https://www.w3schools.com/colors/colors_names.asp

Border colours can be set with these example syntax

NAMED COLOURS:

```
<body>
  <h1 style="border:2px solid pink ;">Hello World</h1>
  <h1 style="border:2px solid cyan;">Hello World</h1>
  <h1 style="border:2px solid purple;">Hello World</h1>
</body>
```

RGB – Red Green Blue Model

```
<body>
  <h1 style="background-color:rgb(255, 0, 0);">rgb(255, 0, 0)</h1>
  <h1 style="background-color:rgb(0, 0, 255);">rgb(0, 0, 255)</h1>
  <h1 style="background-color:rgb(60, 179, 113);">rgb(60, 179, 113)</h1>
  <h1 style="background-color:rgb(238, 130, 238);">rgb(238, 130, 238)</h1>
  <h1 style="background-color:rgb(255, 165, 0);">rgb(255, 165, 0)</h1>
  <h1 style="background-color:rgb(106, 90, 205);">rgb(106, 90, 205)</h1>
```

```
</body>
```

HEX - Hexadecimal

```
<body>
  <h1 style="background-color:#ff0000;">#ff0000</h1>
  <h1 style="background-color:#0000ff;">#0000ff</h1>
  <h1 style="background-color:#3cb371;">#3cb371</h1>
  <h1 style="background-color:#ee82ee;">#ee82ee</h1>
  <h1 style="background-color:#ffa500;">#ffa500</h1>
  <h1 style="background-color:#6a5acd;">#6a5acd</h1>
</body>
```

HSL – Hue, Saturation, Lightness

```
<body>
  <h1 style="background-color:hsl(0, 100%, 50%);">hsl(0, 100%, 50%)</h1>
  <h1 style="background-color:hsl(240, 100%, 50%);">hsl(240, 100%, 50%)</h1>
  <h1 style="background-color:hsl(147, 50%, 47%);">hsl(147, 50%, 47%)</h1>
  <h1 style="background-color:hsl(300, 76%, 72%);">hsl(300, 76%, 72%)</h1>
  <h1 style="background-color:hsl(39, 100%, 50%);">hsl(39, 100%, 50%)</h1>
  <h1 style="background-color:hsl(248, 53%, 58%);">hsl(248, 53%, 58%)</h1>
</body>
```

CSS – CASCADING STYLE SHEETS

CSS is used to control the format and layout of a webpage. This formatting includes the ability to control the colour, font, size of text, spacing between elements, positions of elements, background images and their relevant positions and sizes, and much more.

The term ***cascading*** is used to mean that any style applied to a parent element will also apply to all children elements within the parent. For example, if the colour of the body text is set to red, all text (headings, paragraphs, etc) within the body, will be red unless explicitly specified otherwise.

CSS can be added to HTML documents in 3 ways:

- Inline – using the **style** attribute inside the HTML elements
- Internal – using a **<style>** element in the **<head>** section of the HTML document
- External – using a **<link>** element to link to an external style sheet

INLINE CSS

This is used to apply a unique style to a single HTML element by using the **style** attribute.

```
<h2 style="color:yellow;">A yellow heading</h2>
```

INTERNAL CSS

This is used to define a style for a single HTML page and is defined in the **<head>** section of the page within an **<style>** element. The stated style will apply to any content with the specified elements.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>
    body {background-color: gray;}
    h1 {color: blue;}
    p {color: red;}
  </style>
</head>
<body>
  <h1>This is a heading</h1>
```

```
<p>This is a paragraph.</p>
</body>
</html>
```

EXTERNAL CSS

This can be used to define styling for many HTML pages (usually within a website). The link to the sheet must be added to the **<head>** section of the pages. The external style sheet can be written in any text editor and named with a **.css** extension in the format **sheetname.css**.

The web page containing the link to the style sheet will look like this:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <link rel="stylesheet" href="linda.css">
</head>
<body>
  <h1>This is a heading</h1>
  <p>This is a paragraph.</p>
</body>
</html>
```

And the style sheet (for example, linda.css) will look like this:

```
body {
  background-color: orange;
}
h1 {
  color: blue;
}
p {
  color: red;
}
```

Other properties that can be added include

- Font-family
- Font-size
- Border
- Padding
- Margin
- Etc

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>
    h1 {
      color: blue;
      font-family: Lucida Sans;
    }

    p {
      color: violet;
      font-size: 250%;
      border: 3px solid red;
      padding: 15px;
      margin: 40px;
    }
  </style>
</head>
<body>
  <h1>This is a heading</h1>
  <p>This is a paragraph</p>
</body>
</html>
```

FAVICON

A favicon is a small image displayed next to the page title in the browser. Any image can be used, and customised favicons can be created on sites like <https://www.favicon.cc>.

For easy accessibility, add the favicon image to the root directory, then add a **<link>** element to the webpage after the **<title>** element.

!!! STUDENTS TO IMPLEMENT!

TABLES

Tables are used to arrange data into rows and columns. A table is indicated using the `<table>` `</table>` tag. The intersection of a row and a column is called a cell. Cells contain the data items to be displayed, and they are defined with the `<td>` and `</td>` tags. Each table row is defined with `<tr>` `</tr>` and the number of cells defined under a row will indicate the number of columns in that row. Table headers can be defined using the `<th>` `</th>` tags.

```
<body>
  <table>
    <tr>
      <th>Name</th>
      <th>Gender</th>
      <th>Age</th>
    </tr>
    <tr>
      <td>Kofi Manu</td>
      <td>Male</td>
      <td>34</td>
    </tr>
    <tr>
      <td>Adwoa Mansah</td>
      <td>Female</td>
      <td>28</td>
    </tr>
    <tr>
      <td>Elijah Micah</td>
      <td>Male</td>
      <td>51</td>
    </tr>
  </table>
</body>
```

OTHER TABLE TAGS

- `<caption>` - defines a table caption
- `<colgroup>` - specifies a group of one or more columns in a table format
- `<col>` - specifies column properties for each column within a `<colgroup>`
- `<thead>` - groups the header content in a table
- `<tbody>` - groups the body content in a table
- `<tfoot>` - groups the footer content in a table

TABLE BORDERS

Styling can be applied to a table to obtain different styles and shapes.

Example: To apply a style to the table, the header and the table data, formatting can be implemented like this:

```
table, th, td {  
    border: 1px solid black;  
}
```

This structure will give individual borders around the table. To fix this, set the `border-collapse` property to `collapse` to make the borders collapse into a single border:

```
table, th, td {  
    border: 1px solid black;  
    border-collapse: collapse;  
}
```

COLOURING A TABLE

It is possible to set the background colour of each cell and give a white border colour so as to get the impression of an invisible border. Under the styling, we can set the following:

```
table, th, td {  
    border: 1px solid white;  
    border-collapse: collapse;  
}  
th, td {  
    background-color: cyan;  
}
```

ROUND TABLE BORDERS

Rounded borders can be set using the `border-radius` property.

```
th, td {  
    border: 1px solid black;  
    border-radius: 10px;  
}
```

Include the **table** to the CSS selector to include a border around the table.

BORDER STYLES

The **border-style** property can be used to set the appearance of the border.

e.g.:

```
th, td {  
    border-style: dotted;  
}
```

The following values are allowed:

- dotted
- dashed
- solid
- double
- groove
- ridge
- inset
- outset
- none
- hidden

The border-color property can also be used to set the border's colour.

SIZING TABLES

Tables can have different sizes for each column, each row, or the entire table. The **style** attribute is used with the **width** or **height** properties to specify the size of a table, row or column.

!!!STUDENTS TO READ AND IMPLEMENT

PADDING AND SPACING

Tables can adjust the padding inside the cells and also the space between the cells. Cell padding is the space between the cell edges and the cell content. By default, the padding is set to 0. The **padding** property is used to add padding, and the value set is indicated in pixels.

```
th, td {  
    padding: 20px;  
}
```

Padding can be indicated for a particular side of the content. The properties to use are **padding-top**, **padding-bottom**, **padding left** and **padding-right**.

```
th, td {  
    padding-top: 10px;  
    padding-bottom: 20px;  
    padding-left: 30px;
```

```
padding-right: 40px;  
}
```

CELL SPACING

By default, the space between cells is set to 2 pixels. The **border-spacing** on the table element is used to set this property.

```
table {  
border-spacing: 30px;  
}
```

!!!STUDENTS TO READ AND IMPLEMENT

- Colspan and Rowspan
- Zebra-stripping tables (colouring)
- Hoverable tables
- Colgroup